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Design and development of an Autonomous Self-Driving RC Car for the Emirates Robotics Competition in 2023: with emphasis on Hardware and Software

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Abstract

An ongoing challenge encountered by contemporary engineering educators pertains to the effective cultivation of students' enthusiasm towards the fields of science and engineering. One of the most effective strategies to address this challenge involves the implementation of a technological competition that encompasses elements of publicity, technology, and student engagement. This study investigates the construction of an autonomous self-driving 1/10th scale remote-controlled car. The objective of the project was to actively engage in and emerge victorious in the Emirates Robotics Competition of 2023, while enabling the research into suitable technologies for its uses in urban autonomous vehicles. The RC car platform implemented in this study utilizes the Waveshare JetRacer Pro AI Kit B, which is built upon the NVIDIA Jetson Nano platform. The RC car platform underwent customization in accordance with the specific requirements of the competition, resulting in its ability to execute rapid and secure maneuvers. Autonomous driving systems encompass a range of essential functions, including obstacle detection, traffic sign detection, road following, and vehicle control. These functions are achieved through the integration of various pre-existing algorithms that have been appropriately modified to meet the specific requirements of autonomous driving. This paper presents a comprehensive analysis of the hardware and software components of the RC car, as well as an in-depth examination of the deep learning architecture and effective modifications made to the existing algorithms, which will serve grounds for better navigation in live environments that modern vehicles are subjected daily.

Keywords— autonomous racing, autonomous car, ROS, CNN, DNN

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1. Introduction

Autonomous self-driving vehicles (AVs) represent a burgeoning phenomenon that has already exerted a significant influence, precipitating a revolution within the realms of transportation and technology. Autonomous Vehicles (AVs) [1]-[3] have emerged as a tangible technological advancement and are demonstrating their efficacy across diverse applications. The implementation of autonomous taxis and shuttles on public roads has been observed in various countries, and these vehicles have been well-received by customers [1]-[5]. Additionally, autonomous heavy-duty class-8 trucks have demonstrated their effectiveness in long distance trials, gaining the trust of fleet owners [6][7]. Autonomous mining dumpers have also proven their capability to operate in rugged and harsh environments [5]. Furthermore, autonomous tractors have made significant contributions to sustainable agriculture [9]– [11]. The United Arab Emirates (UAE) has already demonstrated its endorsement of the advancement of driverless technology [12]– [14], highlighting numerous benefits including decreased environmental pollution, accidents, and traffic congestion. The Dubai Metro system serves as a notable illustration of autonomous public transportation [15], [16]. The public transportation system in Dubai is undergoing a comprehensive transformation as a result of the remarkable achievements of autonomous operational systems. The Roads and Transport Authority (RTA) plans to implement autonomous taxi services in the Emirate by the year 2023, with a projected fleet of 4,000 units by 2030 [17], [18]. Dubai will be the inaugural city, outside of the United States, to implement the utilization of such vehicles. In order to enhance the acceptance of autonomous vehicles (AVs) and prepare the United Arab Emirates (UAE) for the future, competitions such as the Emirates Robotics Competition 2023 (ERC'23) are of utmost importance [19]. The objective of ERC'23 is to motivate undergraduate students from nearby universities to engage in addressing contemporary societal and industrial issues through the exploration of robotic challenges. The ERC'23 conference was collaboratively organized and financially supported by the Rochester Institute of Technology (RIT) [21], Dubai Future Labs (DFL) [14], and Khalifa University [22]. The ERC'23 competition consists of two distinct categories of challenges, namely manipulation and navigation. The team in question actively engaged in the navigation challenge, with further information available in references [20] and [23]. The visual representation of the track can be observed in Fig. 1. This paper will cover related work that support the use of neural networks specifically for this Autonomous RC car competition, thorough methodology for the hardware, software, and competition specific details taken into account for the parameters used, and finally cover the results and conclusive details of this paper.

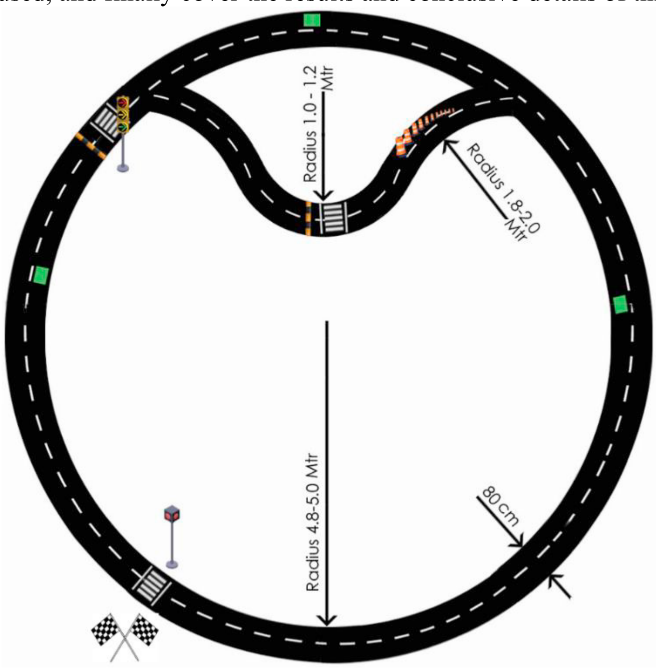

Fig. 1. Challenge track with obstacles, traffic signals, pedestrian crossing, humps and working area safety cones.

2. Related work

In addressing the navigation challenge, extensive investigation was conducted in both the software and hardware domains. Within the present context, a comprehensive software pipeline that encompasses the entire process, utilizing a data-driven approach such as a convolutional neural network (CNN) [22], has been employed (as shown in Fig. 2(c)). The conventional sequence of perception, planning, and control is substituted by an end-to-end software pipeline, which employs a neural network to directly translate sensor input into control output [24]–[28]. The objective of the partial end-to-end approach is to substitute or merge modules using a deep neural network (DNN), as illustrated in Fig. 2(b). One advantage of employing a DNN is its ability to generate an intermediary, lower-dimensional depiction of the racetrack, such as a trajectory. This representation can subsequently be utilized by traditional control systems, such as a proportional-integral-derivative (PID) controller (as shown in Fig. 2(a)). The ultimate output of the actuator (specifically, the steering angle and throttle position) is not explicitly forecasted, as opposed to a comprehensive end-to-end system. While autonomous racing provides an optimal setting for the evaluation and validation of end-to-end methodologies, it is important to consider certain factors that may impact the effectiveness and reliability of these methodologies:

i. Specified drivable area.
ii. Limited signages.
iii. Limited class of objects.
iv. Clear objective for training (fastest lap time).

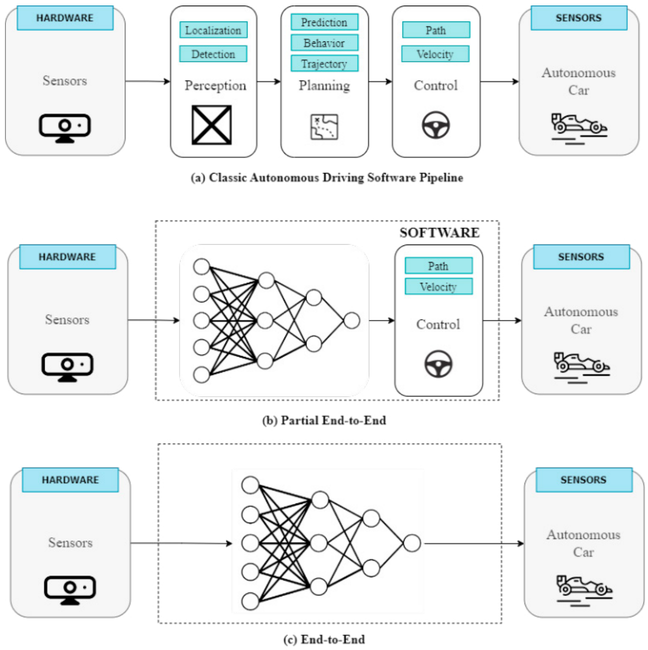

Fig. 2. Classic Autonomous driving pipeline, (b) Partial End to End pipeline, (c) End to End pipeline.

The primary challenges in end-to-end pipelined systems, similar to other DNN applications, pertain to their overall applicability and efficiency [29]– [31]:

i. Difficult to learn the vehicle dynamics parameters- especially the non-linear vehicle and tire dynamics.
ii. There will always be a reality gap.
iii. A large variety of data is required for model training.
iv. Instances of extreme failure can arise from events that fall outside the distribution of the model. For example, driving at high speeds poses a challenge as the model may not possess sufficient preparedness to respond accurately.

In the below Table I, end to end approaches for autonomous racing on hardware platforms have been summarized.

3. Methodology

The study utilized the pre-configured setup (Jetson image-jetcard [41]) provided by Waveshare JetRacer Pro AI Kit B [42], which is built upon NVIDIA's Jetson Nano platform. The primary rationale for utilizing the Jetson Nano, over similar computing platforms such as the Raspberry Pi4, stems from its incorporation of a 128-core NVIDIA Maxwell GPU. The use of the Jetson Nano platform provides a suitable environment for local processing of deep learning on-board the car, which would otherwise be challenging on other general purpose GPUs. The jetcard package includes several essential software components, namely PyTorch, Torchvision, TensorFlow, Traitlets, Jetcam, TorchtRT, and JetRacer. PyTorch and Torchvision are utilized for deep learning tasks, while TensorFlow is employed for machine learning purposes. Traitlets is employed to establish and sever the connection between the camera image and the text. Jetcam is responsible for camera-related functionalities. TorchtRT is utilized to convert PyTorch models into TensorRT models, which enables faster inference. JetRacer is an additional board that interfaces with the motors and provides a 1/10th-scale autonomous RC car. This car effectively captures scaled dynamics, sensing modalities, and decision-making capabilities, making it a safe and accessible platform for research and development. The sensor suite of the car includes a CMOS camera and an inertial measurement unit, while the actuators are controlled by an electronic speed controller. It is worth noting that not all the features of the package were utilized. The hardware illustrative block diagram has been shown in Fig. 3.

TABLE I. OVERVIEW OF END-TO-END APPROACHES FOR AUTONOMOUS RACING

References	End-to End approach	Method
Lee et al. [32]	Vision-based planning using deep learning	Model Predictive Control (MPC), CNN
Tatulea et al. [33]	Trajectory planning using deep learning	Nonlinear-MPC (NMPC), DNN
Drews et al. [34]	Localization using deep learning	CNN, Long Short-Term Memory (LSTM), MPC
Mahmoud et al. [35]	Vision-based planning using deep learning	CNN, LSTM
Chisari et al. [36]	Vision-based planning using reinforcement learning	Soft Actor Critic (SAC), policy output regularization
Lee et al. [37]	Vision-based planning using reinforcement learning	Bayesian decision making
Pan et al. [38]	Vision-based planning using reinforcement learning	Imitation learning
Cai et al. [39]	Vision-based planning using reinforcement learning	Imitation learning
Brunnbauer et al. [29]	Vision-based planning using reinforcement learning	Model based reinforcement learning
Ivanov et al. [40]	Reinforcement learning	Variation of algorithms

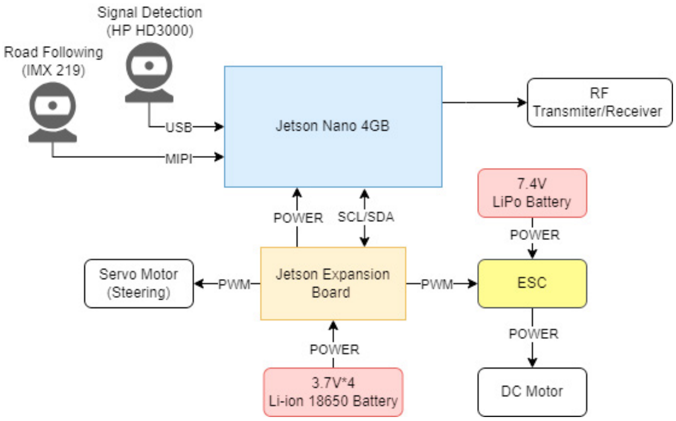

Fig. 3. Hardware block diagram

To make the algorithm simple, two cameras have been used. The CMOS camera (IMX 219) [41] provides a wide field of view, but at the same time, image straightening would have been required due to the irregularities towards the boundary of the image. Therefore, CMOS camera have been used for racetrack tracking purpose (as shown in Fig. 4), and during training the model, images from CMOS camera were taken in such a manner that it covers maximum amount of the track in the image, so that image straightening can be avoided (as shown in Fig. 5).

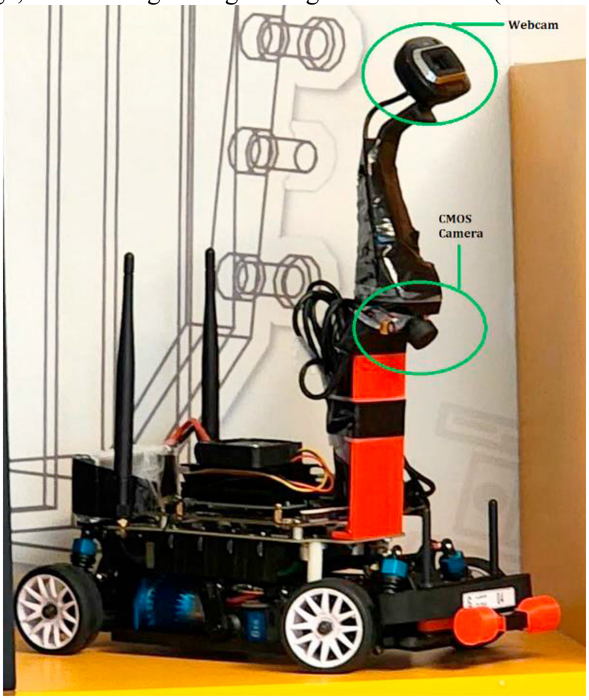

Fig. 4. Scaled RC car used for autonomous racing.

Further to capture the traffic signal information another webcam (kept at an inclined angle, as shown in Fig. 4) has been used. Further keeping traffic signal and racetrack in the image is adding a lot of noise, and the resultant model is not working properly (as shown in Fig. 5). The Flowchart of the algorithm has been shown in Fig. 6.
To accomplish this, a separate frame has been identified within the default frame. The centre of the new frame coincides with the default frame. When the traffic signals are detected in the new cropped frame, the prediction value-based classifier will execute and identifies the appropriate action for the signal depending on threshold value (0.9).

Training a neural network is the process of transferring knowledge from one form to another, namely, from an intended desired behaviour to neural network weights and biases. Behavioural planning has been implemented. The aim is to prepare a model that can help in safe manoeuvring, and for the same reason, the following approaches were considered:

i. The position of the autonomous RC car on the track was trained for extreme conditions like zig- zag motion. So that the intent of the vehicle always remains at the center of the track.

ii. As speed was fixed and there was no throttle control in place, during curvatures to avoid obstacles, a safe distance from the obstacle was the prime objective, and therefore different scenarios were tested for the same.

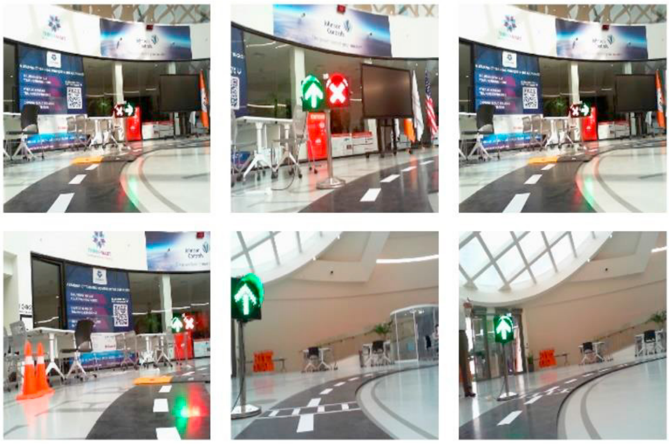

Fig. 5. Actual scene from the racetrack

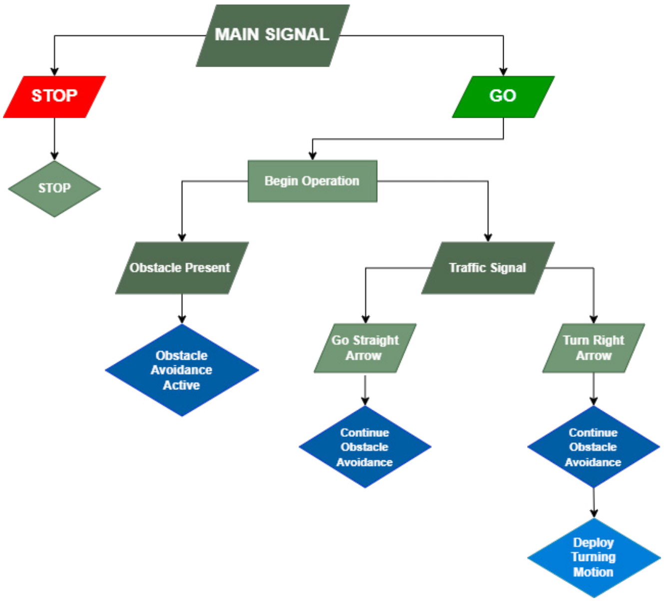

Fig. 6. The flowchart of the algorithm used in ERC'23

The goal is to reduce the lateral and heading errors to stay as close to the reference line (in this case, the dashed line at the center of the track), and to maximize the safe velocity (and avoid drifting) of the RC car without hitting any obstacles. To achieve this, the model used was trained for extreme situations and velocity checks were done via hit-and-trial methods. In the current scope, a full end-to -end software pipeline that is data driven approach like a convolution neural network (CNN), has been used.

Initial trial was performed on the printed track, received from waveshare kit [42]. To train the model, 200 images of the track in various orientation around the track by following it were taken. Model was saved using 25 epochs, with the following adjusted values:

i. STEERING_GAIN = 1.18
ii. car.steering_offset = 0.2
iii. car.throttle = 0.3
iv. car.throttle_gain = 1

For the competition track, ResNet-18 was used over deeper architectures due to faster training times and less parameters needing to be changed (steering bias, steering gain, and throttle position), ideal for use for local computation. Two models were developed using ResNet-18 architecture [44]:

i. Model 1 (Moving around the outer circle): Like the trial run, in this case 2000 images were used in modelling for various scenarios. Then ran the RC car and adjusted the values for the track. Since it was a right turning circle, following adjustments were taken into consideration:
   a. Steering bias to the right
   b. Less steering gain
   c. Higher throttle.
ii. Model 2 (Moving the inner half circle): In this case 1000 images were used in modelling for various scenarios, with the following adjustments:
   a. High steering gain (for sharper turns)
   b. Low throttle

To tackle the challenges on the track, following considerations were made:

i. Humps: These are placed at the zebra crossing areas with a height of $40\ mm$ and a width of $39\ cm$. The ground clearance of the RC car after placing the on-board accessories (camera, batteries, controller etc.) is $8.5\ mm$ and the front bumper of the RC car was initially getting stuck with the inclined angle of the hump. This issue was resolved by using a 3D printed rotary mechanism, shown in Fig. 7(a). It helps in the initial lift of the RC car, during the starting of the hump inclination which later been addressed by the front wheels of the RC car.
ii. Traffic signaling panels: Signaling panels were kept simulating a realistic traffic scenario where vehicles must obey these signals to stop, follow straight ahead, and make a turn (right side) (as shown in Fig. 7(b)(c)). To address the challenge, various images for each category for different orientations were captured to prepare a prediction value-based classifier model. The predicted value for the classifier model works in the following manner:
   a. If follow straight ahead signal is above threshold value (0.9), the RC car will run on model 1 which was trained to go around the outer track.
   b. If the predicted value for turn (right) signal is higher than threshold, the RC car will run on model 2 which was trained to take a right when found the opportunity to.
   c. If the prediction value for stop was above threshold value, car throttle was set to 0.
   d. For predicted values less than threshold value, RC car will keep on following its current state.
iii. Safety Cones: these are placed in the inner semicircle part of the track. They have a base of $19\ cm * 17\ cm$ and a height of $30\ cm$, as shown in Fig. 7(d). These were addressed during the training of Model 2 by training the model for obstacle avoidance.
iv. Obstacles: These are rectangular green boxes with a $15\ cm * 30\ cm$ base and a minimum height of $10\ cm$ (as shown in Fig. 7(e)). These were addressed during the training of Model 1 and 2 by training the model for obstacle avoidance also.

4. Results

The model presented in this study demonstrates the ability to achieve satisfactory performance on the track, resulting in the attainment of the first prize in the challenge. The complete deployment videos and code can be accessed at the following sources: [31][44][46]. During testing and the final competition, it was evident that ideal light conditions were needed to ensure stable prediction of the target. In order to effectively build on this platform for further research, a larger dataset for most possible lightings would increase the reliability and accuracy. Using sensors like a 3D lidar would further provide better close range depth perception which can further improve the effectiveness of the AV when in motion. The increase in dataset size and sensor paths will require a larger platform which would further increase the scope of research for better understanding various environments that AVs are subjected to. Therefore, the current limitation of this study is the relatively small sample size of algorithms included and the physical platform limitations, which hinders the potential for future research. Nonetheless, the system presented in this work serves as a robust foundation for subsequent investigations into AV systems using training data captured through visual perception.

(a) Hump at zebra crossing area.

(b) Traffic signal at the departure area.

(c) Traffic signal at junction area.

(d) Working area safety cone.

(e) Rectangular green box obstacle.

Fig. 7. Traffic related items

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